POKHARA UNIVERSITY

Level: Bachelor　　Semester: Fall　　Year: 2025

Programme: BCSIT　　Full Marks: 100

Course: Object Oriented Analysis and Design　　Pass Marks: 45

Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section "A"

Very Short Answer Questions

Attempt all the questions. [10×2=20]

1. List down the benefits of OOAD in system development.
2. What are the two key aspects of requirement?
3. What do you mean by domain modeling?
4. Write any two differences between <<include>> and <<extend>> relationships.
5. Differentiate between aggregation and composition.
6. Define pattern. List out GRASP pattern.
7. What is Open Closed Principle (OCP) in SOLID principles?
8. Who are the "Gang of Four" (GoF) in the context of software design?
9. What is adapter pattern in GOF patterns?
10. Why do we perform "Noun Phrase Analysis" during the domain modeling phase?

Section "B"

Descriptive Answer Questions

Attempt **any five** questions. [5×10=50]

11. a. Define association, aggregation, and composition with UML notation.
    b. Explain the process of Requirement Elicitation. Describe three different techniques used to gather requirements from stakeholders.
12. What are the behavior diagrams? Compare use-case diagram and activity diagram highlighting their purpose and importance in OOAD.
13. Explain the components of Activity Diagram with their UML notations. Using UML, draw a detailed activity diagram considering given scenario. A customer purchases a product from an e-commerce website in Nepal and chooses eSewa as the payment method. The customer logs in to eSewa, reviews the payment details, and authorizes the payment. The eSewa system verifies the user credentials and checks the account balance. If the payment is successful, eSewa confirms the transaction, the e-commerce system records the payment, and the order is placed successfully. If the payment fails due to insufficient balance, incorrect credentials, or transaction timeout, the payment is cancelled and the customer is notified.

14. Create a detailed class diagram for a library management system, incorporating generalization, dependency, constraints, composition and aggregation for database management.
15. Explain in details about General Responsibility Assignment Software Patterns (GRASP).
16. a. Explain high cohesion principle of GRASP with suitable examples.
    b. What problem does Adapter Pattern solve? Explain in detail.

Group "C"

Long Answer Questions

Attempt any two questions. [2×15=30]

17. Explain the <<include>>, <<extend>>, and Generalization relationships in detail. Draw a Use Case Diagram for an "Online Banking System" that demonstrates all three relationships.
18. Explain SOLID principles in object-oriented design. Discuss why following these principles is important for building flexible, maintainable, and reusable systems.
19. What is activity diagram? Draw a activity diagram with Swimlanes for the given scenario:
    You are tasked with designing the workflow for a secure ATM system that coordinates actions between a Customer, the ATM Machine, and the central Bank Database. Using a UML Activity Diagram with Swimlanes, model the complete withdrawal process. The workflow begins when the customer inserts their card and enters a PIN. The ATM must first send this data to the Bank Database for authentication. If the PIN is invalid, the ATM should display an error message and immediately eject the card, terminating the session. If the PIN is valid, you should be able to withdrawal the cash and print the receipt.